Substrate specificity prediction of transporters using triplet loss and protein language models

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Context

Transmembrane transport proteins:

- Crucial for substrate transport and metabolic pathway recreation
- **Experimental curation limitation** makes dataset:
- **Sparse** and **imbalanced**
- **4,455 non-identical sequences**
- Transporting **96 substrates**
- **One majority** class(1,226 samples)
- **Long-tailed minority** classes (<10 samples)

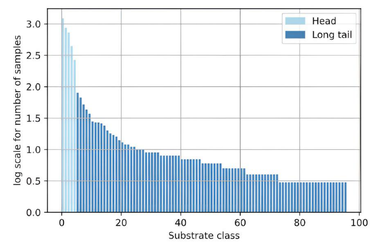


Figure 1: log-scale of sample distribution on each class

Requires a **computational model:**

- To predict specific substrates of a transport protein
- Effective training on sparse, imbalanced dataset

Approach

Machine learning model:

- **Predict** the specific substrate of a transport protein
- Employ **triplet loss function** and **sampling strategy**
- Modify **embedding space** of pretrained ProtBERT-BFD
- Use **K-NN algorithm** to predict candidate substrate
- Effectively train **sparse classes** with as few as three samples

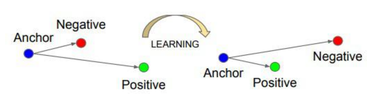
Methodology

Protein Language Model:

- ProtBERT-BFD: 420M parameters
- 30 layers with 16 attention heads
- Trained on BFD-100 dataset with 393B amino acids
- Mask Language Modeling (MLM) self-supervised pretraining task [1].

Triplet sampling:

- *a, n, p:* **anchor**, **negative**, and **positive** samples
- Every training is **anchor** once
- **Positive: random** sample with **same label**
- **Negative: random** sample with **different label**



Triplet loss function:

- **Minimizes intra-class distances** while **maximizing the inter-class distance** during training.
- Defined for all possible sets of triplets in the training set [2]
- $f(x_i)$ shows the representation of the sample x_i
- α is a margin enforced between positive and negative pairs

$$L = \sum_i^N [||f(x_i^a) - f(x_i^p)|| - ||f(x_i^a) - f(x_i^n)|| - \alpha]$$

Training:

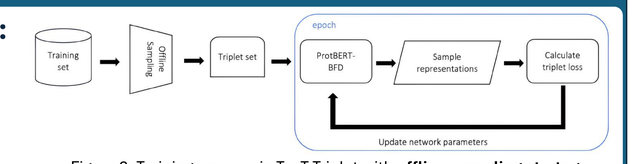


Figure 2: Training process in TooT-Triplet with **offline sampling strategy**

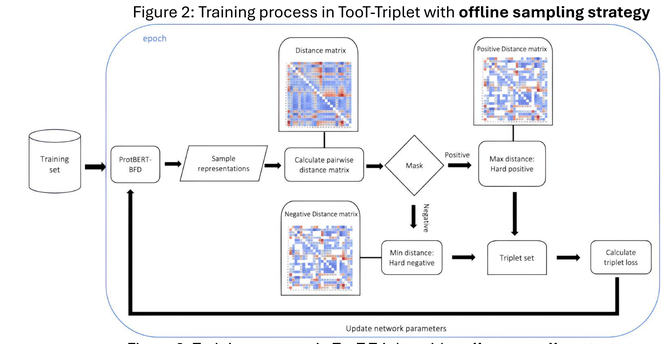


Figure 3: Training process in TooT-Triplet with **online sampling strategy**

Results

Table1: 3-fold Cross-validation results for TooT-Triplet using Manhattan distance

Folds	Accuracy	Precision	Recall	F1-Score	MCC
Fold-1	0.997	0.875	0.718	0.700	0.852
Fold-2	0.998	0.910	0.756	0.712	0.893
Fold-3	0.998	0.897	0.754	0.722	0.878
Average	0.998	0.896	0.742	0.711	0.874
±SD	±5.77e-04	±2.05e-02	±3.21e-02	±3.31e-02	±2.45e-02

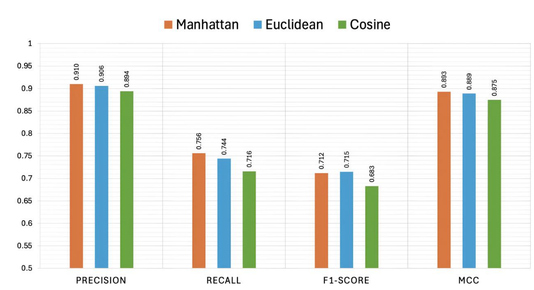


Figure 4: TooT-Triplet results in independent test set

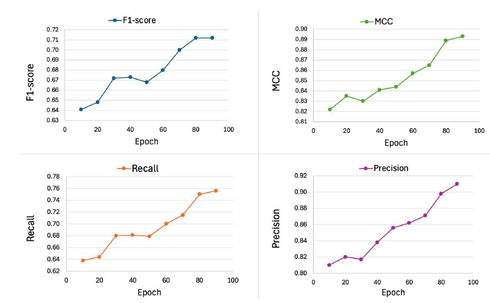


Figure 5: Metric evaluation in TooT-Triplet training process using Manhattan distance

Table2: Results for online vs offline sampling strategy for independent test set

Sampling strategy	F1-score	MCC
Online sampling	0.680	0.835
Offline sampling	0.712	0.893

Discussion

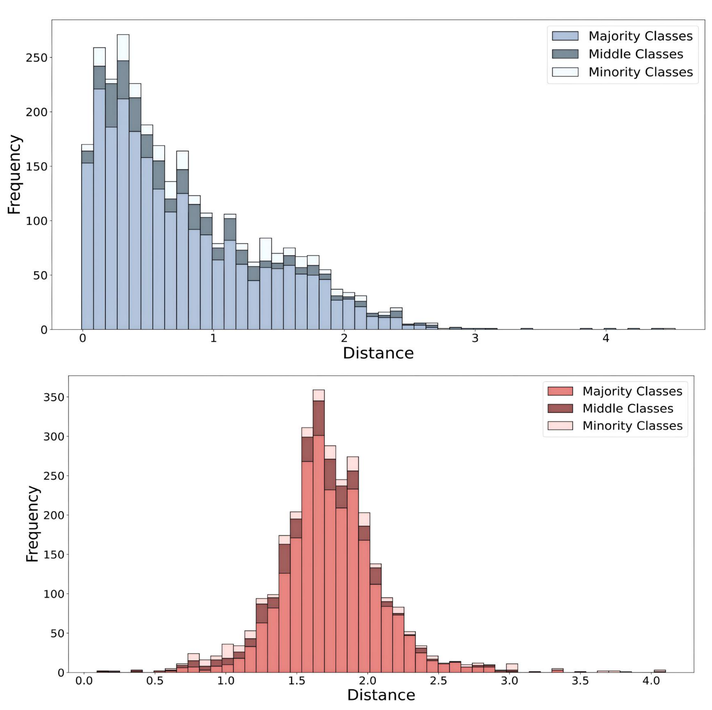


Figure 6: Distribution of sample distances for **positive (top)** and **negative (bottom)** nearest neighbor with offline sampling strategy

Best model:

- offline sampling Manhattan distance
- Optimizer: AdamW
- Learning rate: 1e-5
- Epochs: 91

- k-NN algorithm in high dimensions depends on **distance metric.**
- Overall, **positive samples** have **lower average distance** than negative.
- **Increasing silhouette score** with size of class indicates **consistency of samples in each class**.

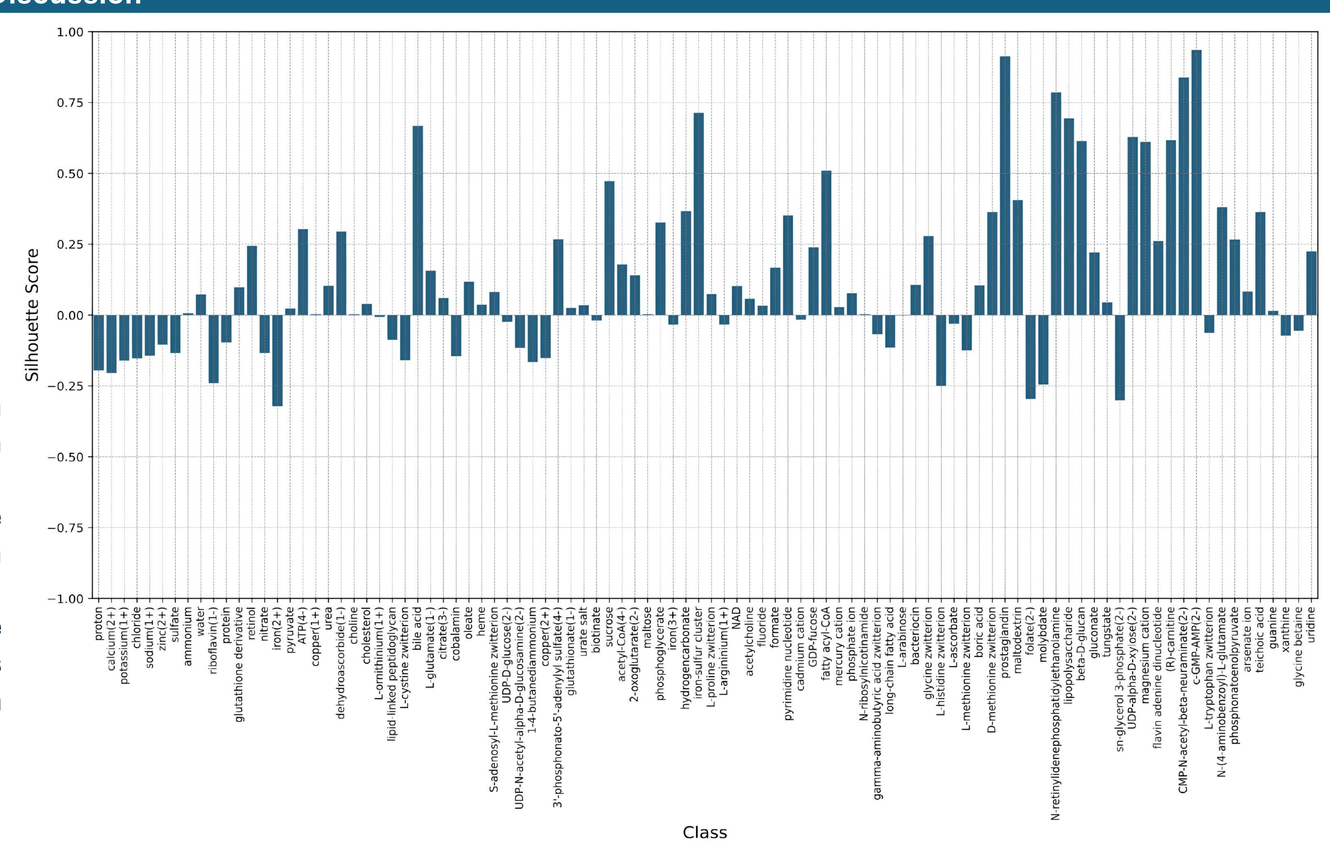


Figure 7: Silhouette score for each class in training set sorted by size of the classes

Conclusion

TooT-Triplet model:
Predicts **specific substrates**
91 classes in the dataset
MCC = 0.89

- **One-shot classification**
- Offline **triplet sampling**
- Manhattan distance

- Provides **the top 3 substrates**
- Ranked by **distance**
- As **reliability measure**

References

[1] Elnaggar, Ahmed, et al. "Prottrans: Toward understanding the language of life through self-supervised learning." *IEEE transactions on pattern analysis and machine intelligence* 44.10 (2021): 7112-7127.

[2] Schroff, Florian, Dmitry Kalenichenko, and James Philbin. "Facenet: A unified embedding for face recognition and clustering." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2015.